National University of computer and emerging sciences

FINAL YEAR PROJECT PROPOSAL
BS-CS FALL 2021

EVENTIVE
(Carpooling System)

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Project Details

1.1. Project Area of Specialization

App Development and ML.

1.2. Project Timeline

Project Start Date	**Project End Date**
September 2022	April 2023

1.3. Project Summary

Carpooling is one of the latest technologies discovered which has made travelling convenient and efficient to the common man. It also known as car-sharing in which one can travel to their destination while sharing the vehicle and the expenses incurred hence reduce single individuals travelling cost, fuel cost and tiredness. Due to the growth in the population, there is inadequate transportation through personal vehicles which gives rise to traffic congestion on roads and damages the eco-friendly environment by increasing air and noise pollution. Hence carpooling is the optimal solution to solve these problems. we will make an Android based application that will allow passengers to collaborate with other like-minded people and plan out their journey using the easy UI of the app after signing in to it.

The main aim is to provide a type of mobility service that, through a joint digital channel enables users to plan, book, and pay for multiple mobility services through carpooling. The concept describes a shift away from personally-owned modes of transportation and towards mobility provided as a service. The key concept is to offer travelers run time solutions based on their travel needs. This carpooling system shows that the user can get from one destination to another by using a combination of multiple rides. For e.g., if a user wishes to move from a point A to point C but fails to find a ride connecting the two points directly, our system will generate a combination of rides operating on that route, which the user can book simultaneously in order to reach its destination through a combination of carpools. In addition, the user can choose their preferred trip based on cost, time, and convenience. At that point, any necessary bookings would be performed as a unit. It is expected that this service should allow roaming, that is, the same end-user app should work in different cities. Furthermore, an individual's safety is also an important aspect in this scenario therefore this system provides gender specific carpooling and live tracking system for rides. Together with other emerging vehicular technologies such as connected cars and electric vehicles, carpooling is contributing to a new type of future mobility, which is autonomous, connected and shared.

1.4. Project Objectives

This Carpooling system aims to provide one stop solution to daily commute problem without any hassle, making it more flexible and reliable while creating great connections with the help of an e-platform where commuters come together to share pleasant journey. Carpooling is a revolutionary and a unique way of travelling. Whether you own a bike, car or a rider, just post your ride details in our app and we will match you with co-riders on your way. We want to provide a hassle-free transit to our users along with a great experience.

- Enable rider to create events that would specify the following information:
  - The total vacancy in the car.
  - The time at which the event is going to take place.
  - The Final destination.
- Development of the logic that would enable-
  - Poll in the location information of all the intended recipients.
  - Take decision based on the context on the location.
  - Send SMS to all the selected recipients and handle the accepted or rejected messages received from the recipients.
- Generate a tracking system using Google Map that shows the initiator the map between his location, all the recipients that agreed to his car-pooling event and the final destination.
- Reduce overall traffic congestion on the roads and single occupancy car trips by implementing carpooling system.
- Promoting alternative modes of transport.
- Reduce peak hour congestion, pollution and other poisonous gases emission in air.
- Cost effective transit, by sharing the cost of driving one car.
- Reduces driving-related stress for participants
- Provide social connections in the society.

1.5 Literature Review / Background Study

Carpoolyn [1] is Pakistan's 1st Carpooling (Ride-Sharing) App with an aim to make Traveling accessible, affordable, and safe for Everyone Across Pakistan. It connects the drivers with passengers traveling on the same routes to share the fuel cost.
Have A Car? Just charge for empty seats in your car and make money to cover your commuting cost.
Need A Ride? Share a ride with drivers heading towards the same destinations & reduce your traveling cost.

Ridely [2] is rideshare services in Pakistan. It's an emerging app that bring a new concept of route matching in Carpool. This carpooling app allows its users to select rides based on the route match between two cities of Pakistan.

Savaree [3] is a Pakistan-based cab service that offers affordable and safe rides. It offers a carpooling app that allows users to find rides to places they would like to go. It also matches passengers with drivers going along specific routes or to specific events. Savaree was founded in 2014 and is based in Lahore, Pakistan. It also provides inter-city services.

BlaBlaCar [4] is the world's largest long-distance ridesharing community. Conceived in December 2003 by Frédéric Mazzella, and founded in 2006, BlaBlaCar connects drivers and passengers willing to travel together between cities and share the cost of the journey. BlaBlaCar has more than 20 million members across 19 countries. Members must register and create a personal online profile, which includes ratings and reviews by other members, social members show how much experience they have of the service, meaning those with more-known as "ambassadors" - attract more ride shares.

FolksVagn [5] offers a community-based system that helps people share rides with others. While the passengers get rides at costs much cheaper than a regular taxi service, the car owner gets a share of the fare. It is open only to corporate clients as it requires a corporate email for registration and has a prepaid account or online wallet system to pay for the ride.

Taxi for sure [6], the famous taxi-hire application on android platform is the first car sharing application who took the initiative and introduced Carpooling for "Vacationers" i.e. for those who are on vacations and want to spend less on travelling to save their pocket.

Problem:

One major shortcoming of these applications is that all the applications are focusing on connecting_drivers and passengers that have same destination. No application is focusing on connecting riders with dissimilar destination. Our main focus is to solve this issue by using efficient ML algorithms for real-time route matching.

Security is another concern specially for women. In today's world it's hard to rely on anyone. In our project we are focusing on gender specific mechanism of carpooling. In this way, we can create more opportunities for female drivers and passengers.

1.6. Project Implementation

Carpooling system provides the benefits of sharing rides with different peoples. By surveying different papers, it is found that it uses heuristic searching techniques for searching the cars. By using Euclidean distance techniques, it finds the nearest car that has to be assigned to passenger. It usually uses Global Positioning System (GPS) for finding the locations of cars. In study it is found that, Route matching techniques are used in order to match drivers and passenger's locations by using Google maps API, and then based on their route, driver is allocated to passenger. In most of carpooling system, it is observed that the Dijkstra algorithm is used for detecting the shortest route of destination of passenger. Dijkstra algorithm is usually used for detecting the optimal minimum path. It is found that is checks for maximum possibilities and then chooses the minimum route for the journey. Similarly, Dijkstra algorithm is also used to connect nearest vehicle.

1.7. Benefits of Project

- **Save money:** Share the costs of driving with other riders.
- **Reduce stress:** Read, listen to music, or relax when you're not driving.
- **Save time:** As a carpool, you can drive in the HOV (high occupancy vehicle) lane.
- **Help the environment:** Carpooling creates cleaner air and safer communities.
- **Prolong the life of your vehicle:** Shared driving puts fewer miles on your own car.

- **Reduces traffic:** When three people carpools, there are two fewer cars on the road. Automatically traffic on roads reduces.
- **Safety:** Our gender specific approach help female to travel more freely and securely.
- Employer financial tax incentives for supporting carpooling.

1.8. Technical Details of Final Deliverable

- Key Frame Extraction algorithms for CCTV videos streams
- Anomaly Detection of various suspicious activities.
- Aggregation of Anomalous Video Segmentation.
- UI Design so that multiple online/offline streams can be analyzed.

1.9. Final Deliverable of The Project

- An application that provides one stop solution to daily commute problem without any hassle, making it more flexible and reliable while creating great connections with the help of an e-platform where commuters come together to share pleasant journey.

1.10. Core Technology

- React Native, mongoDB, Nodejs, Machine Learning Models, Python.

References

1. E. Ferguson, "Demographics of Carpooling," Transp. Res. Rec. 1496, pp. 142–150, 1990.
2. E. Ferguson, "The rise and fall of the American carpool: 1970–1990," Transportation (Amst)., vol. 24, pp. 349–376, 1997.
3. ONU-Hábitat, "Reporte Nacional de Movilidad Urbana en México 2014-2015," Rep. Glob. en Asentam. Humanos, p. 100, 2015.
4. E. Nechita, G.-C. C. Crişan, S.-M. M. Obreja, and C.-S. S. Damian, "Intelligent carpooling system: A case study for bacău metropolitan area," Intell. Syst. Ref. Libr., vol. 107, pp. 43–72, 2016
5. R. Aissaoui and O. I. Houssaini, "CARPOOLING APPLICATION : KwiGo," no. Apri.
6. Kalogirou, Kostas and Dimokas, Nikos and Tsami, Maria and Kehagias, Dionysis, 2018 IEEE 20th International Conference on High Performance Computing and Communications; IEEE 16th International Conference on Smart City; IEEE 4th International Conference on Data Science and
7. Systems (HPCC/SmartCity/DSS), 1271--1278, (2018)

Project Key Milestones

Elapsed time in (days or weeks or month or quarter) since start of the project	Milestone	Deliverable
Month 1	Practical Analysis of Existing System.	Literature review. Nature of the problem Identify possible challenges. analyze, models, and a logical alternative.
Month 2	Requirements Gathering.	Identify and gather insights of existing systems and get an adequate grasp of the project flow.
Month 3	Proposed Architecture.	Understanding the course of the project. ER Diagram, Data flow diagram, Activity Diagram etc.
Month 4	Prototype 1 + Alpha Testing.	Implementation of components i.e. of user and Driver portals (User login + Driver Registration). Casting Mechanism such as Any Cast. Geofencing.
Month 5	App Development and Promotion Website Development.	Interface development
Month 6	Applying ML/DL Algorithms.	Route Matching, fare calculator, recommendation based on rating, Radius calculator.
Month 7	Integration of Relevant APIs and Modules + Component Testing.	Google map API, Payment API.
Month 8	Testing.	System Testing.